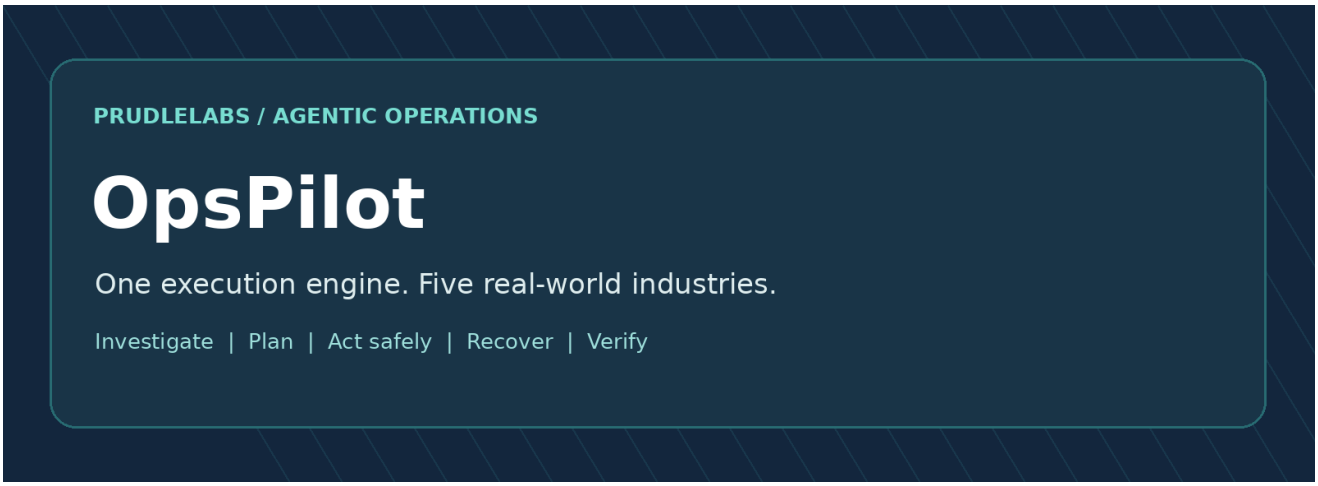


Prudlelabs OpsPilot

Detailed product blueprint, integration specification, implementation roadmap, and Canadian go-to-market playbook

One reusable agentic operations platform • Five industry flagship demos • Realistic business-system simulations

Document status | Working strategy and engineering specification
Prepared for Prudlelabs leadership, solution architects, engineering, Canadian partnerships and sales teams. All demo data is synthetic. Timelines, prices and targets are planning assumptions, not validated customer commitments.

Document control	Value
Version	1.0 — 20 October 2026
Product identity	Prudlelabs OpsPilot (not PoodleOps)
Primary region	Canada — Southern Ontario initially
Target users	Canadian enterprise buyers; Prudlelabs delivery, product, marketing and sales teams
Delivery model	Internal reusable IP first; customer-specific implementations second

How to use this blueprint

This guide serves three groups. Leadership uses the commercial and sequencing decisions. Engineering uses the architecture, interfaces, data models and acceptance criteria. The Canadian sales team uses industry narratives, live demo scripts, objections and pilot offers.

Read first if you are…	Relevant chapters
Managing the business	1–3, 15–19
Designing architecture	4–9, 12–14
Building domain demonstrations	10–13
Selling and delivering in Canada	15–19
Preparing first release	20–22

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1. Executive summary and product thesis

A business problem first. An AI agent second. Verified operational progress always.

Prudlelabs OpsPilot is a reusable engineering foundation for building AI agents that coordinate complex business operations. The customer gives an objective or operational incident. The agent investigates facts across approved business systems, proposes a plan, executes authorized steps, reacts to changes, and checks the result in authoritative records.

Primary business promise

From business exception to coordinated resolution. OpsPilot handles investigation, decisions within guardrails, system updates and follow-up — with traceable evidence and human control.

What we will build

Product asset	Purpose
OpsPilot Core	Case runtime, tool calling, plans, approvals, action records, verification and operator dashboard.
Five industry domain packs	CivicResolve, SupplyResolve, FreightResolve, ClaimResolve, PropertyResolve.
Simulation environment	Stateful mock 311, ERP, TMS, claims, maintenance, communication and scheduling services with fault injection.
Blast Radius Lite	Bounded impact preview computed from current modeled business relationships and policies.
Sales demonstration kit	Eight-minute guided scenarios, controlled exceptions, reset and replay, ROI worksheets and pilot scoping templates.

Non-goals for release 1

- No autonomous legal, municipal, insurance-coverage or claims-settlement decisions.
- No replacement for ERP, TMS, municipal case management or insurance core systems.
- No proprietary tracing platform; consume OpenTelemetry and create business-action evidence.
- No generic low-code multi-agent builder, public app marketplace, or unbounded autonomy.
- No claim that simulated performance proves production ROI or regulatory compliance.

What success looks like

A buyer recognizes the case in the first 30 seconds; the agent queries at least three authoritative simulated systems; a deliberate complication changes its plan; a protected write needs authorization; the result appears in the target system and is independently verified. The sales team can repeat the demonstration from a known reset state.

2. Market fit, buyers and flagship portfolio

Five recognizable stories, aimed at named owners of operational problems.

Canada is moving from isolated AI tools toward integrations and measurable workflow benefits. Statistics Canada reported 19.2% business AI use in Q2 2026. That does not mean those businesses already run autonomous agents; OpsPilot targets organizations ready for operational automation. The City of Brampton publicly listed municipal AI PoC focus areas, creating a useful nearby conversation context, not guaranteed procurement.

Domain pack	Buyer / budget owner	Trigger event	Target outcome
CivicResolve	311 / public works director; municipal CIO	Repeated streetlight complaint	Correct ownership and accepted service action
SupplyResolve	COO / plant manager / procurement head	Critical material delay	Authorized material-recovery plan
FreightResolve	Dispatch director / 3PL operations	Vehicle breakdown	Accepted replacement and updated dispatch
ClaimResolve	Claims operations leader / TPA	Incomplete repair estimate	Evidence complete and adjuster-ready
PropertyResolve	Property manager / facilities head	Water leak and contractor rejection	Safety escalation and confirmed work acceptance

Commercial fit evaluation before a demo

- There is a recurring operational exception, not merely occasional document classification.
- Resolution requires information from at least two separately maintained systems or teams.
- An employee currently checks, chooses and follows up on several possible next steps.
- The organization can provide an accountable process owner, data access and a safe approval policy.
- The operational result can be measured (time, backlog, errors, cost, SLA or customer communication).

Market references: Statistics Canada Q2 2026 business AI adoption; City of Brampton Artificial Intelligence Proof of Concept Program. Sources are listed in Chapter 22.

3. What counts as agentic

Do not rebrand fixed scripts as autonomous business work.

Characteristic	Ordinary workflow	OpsPilot agentic target
Control flow	Steps and branches are predefined	Select investigative steps from available tools within scope
Novel exception	Raise error; human remediates	Reassess evidence, re-plan, seek permission or escalate
Cross-system work	Sequence known APIs	Select relevant systems and reconcile inconsistent answers
Writes	Triggered by workflow conditions	Proposed by agent, validated by deterministic guardrails
Result	API response recorded	Independent business-state and downstream evidence verified

Use LLMs selectively

Agent reasoning is appropriate for reading unstructured reports, selecting a relevant investigation route, comparing business-approved choices and drafting communications. It is not the source of truth for inventory arithmetic, payment limits, staff permissions, SLA dates, duplicate checks or database commits. Those belong in deterministic code and the relevant business system.

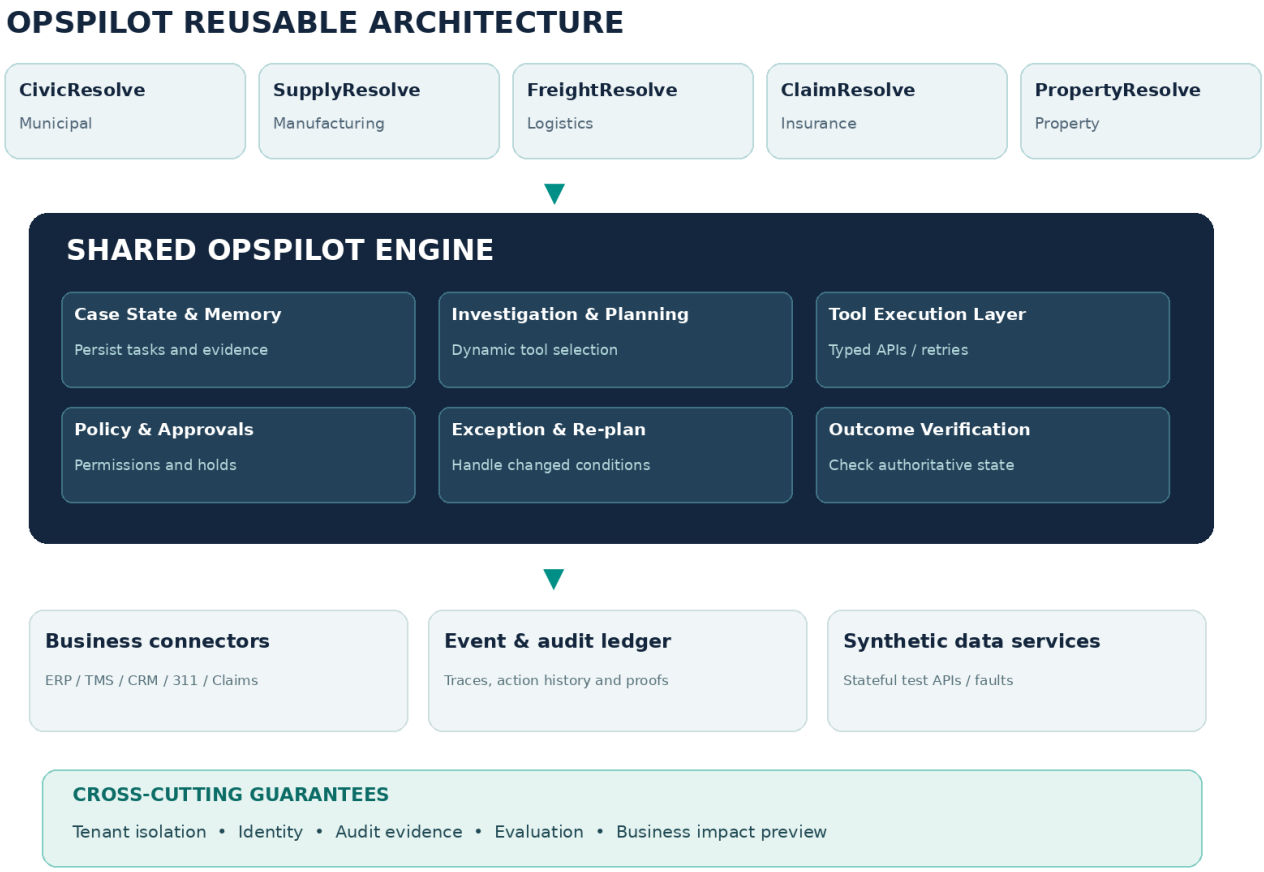
One agent per case by default

Start with a single task agent that plans and calls typed tools. Use specialist services (search, optimization, policy decisions, notifications) as tools, not automatically as additional AI agents. Introduce multiple autonomous agents only if a controlled benchmark shows a benefit.

Design principle
A successful autonomous outcome may be “escalated safely.” The product must prefer a correct escalation over a fabricated completion or an unsafe action.

4. Reference architecture and shared components

The shared runtime is the product; domain packs are configured applications.



Logical services

Service	Responsibility	Implementation choice
Web console	Case queue, timeline, approvals, evidence, replay	Next.js / TypeScript
API and identity	Tenant routing, RBAC, API authentication, audit	FastAPI + OIDC
Case orchestrator	Durable state transitions and interruption/resume	LangGraph checkpoints + background worker
Tool registry	Typed tool descriptions, scopes, versioning, timeouts	Pydantic schemas + registry tables
Policy / impact	Rule evaluation, approval required, related entities	Python rule layer; OPA if justified
Verification	Compare expected versus authoritative actual state	Per-domain verifiers
Business simulators	Deterministic stateful mock APIs, fault injection	FastAPI microservices + PostgreSQL

Service	Responsibility	Implementation choice
Observability	Traces, metrics, action ledger and evaluation	OpenTelemetry + collector + chosen backend

Physical deployment profiles

- Local demo: Docker Compose with platform, PostgreSQL, Redis/worker, mock business systems and optionally local stub model responses.
- Hosted demonstration: isolated cloud environment in an approved region, synthetic records, restrictive access, rate limits and demo reset.
- Customer pilot: use customer-approved APIs and auth; add a customer-hosted connector for restricted networks and sensitive fields; retain human approvals for consequential writes.

5. Case lifecycle and orchestration

A durable business case outlives any single LLM response or API call.



Required case states

State	Meaning	Exit condition
NEW	Event or user creates case	Input validated and owner assigned
INVESTIGATING	Agent gathers and cross-checks facts	Sufficient evidence or unresolved uncertainty
PLAN_READY	Candidate actions with cost / risk / evidence	Policy evaluation completed
WAITING_APPROVAL	Protected action awaiting human authorization	Approve, reject or expire
EXECUTING	Trusted adapter performs authorized action	Receipt returned or timeout classified
VERIFYING	Fresh authoritative state queried	Success, pending, discrepancy or unknown
REPLANNING	New information invalidated the plan	Replacement option or escalation
DONE / ESCALATED	Closed with proof or human handoff	Immutable closeout record

Resume, retries, concurrency

- Persist case_id, thread_id, state_version, plan_version and last authoritative observations.
- Every write proposal carries idempotency_key, expected_version, purpose, expiry and actor identity.
- After a timeout, query current state before retrying; never assume a second call is safe.
- Re-validate authorization and business preconditions just before committing a write.
- Use optimistic concurrency or locking in the business system. A pre-check alone cannot eliminate race conditions.
- Human approval should be tied to a versioned action proposal; if the target state changes, the approval expires.

Recovery principle

For an external business action, reversal may not be possible. Design compensating operations and human-supported correction rather than promising universal rollback.

6. Canonical data model and event contract

Define common case, action and evidence types before adding domain logic.

Entity	Core fields
Tenant	tenant_id, region, roles, settings
DomainPack	pack_id, version, tool_scopes, rules, verifier
Case	case_id, tenant_id, domain, objective, priority, state, owner
BusinessEntity	entity_type, external_id, source_system, current_version
Observation	observed_at, source, fetched_by, data_hash, validity
ActionProposal	proposal_id, case_id, tool, arguments, preconditions, impact
PolicyDecision	decision_id, rule_version, result, reasons, expiry
Approval	approval_id, actor, proposal_version, status, timestamp
Execution	execution_id, idempotency_key, receipt, actual_state
Verification	expected_state, observed_state, evidence_refs, status

Example canonical proposed action

```
{
  "case_id": "case_freight_2084",
  "proposal_id": "act_00031",
  "domain": "freight",
  "tool": "dispatch.assign_replacement_carrier",
  "resource": {"shipment_id": "SHP-2084", "carrier_id": "CAR-44"},
  "preconditions": {"shipment_version": 12, "carrier_available": true},
  "estimated_incremental_cost_cad": 420,
  "idempotency_key": "case_freight_2084:act_00031:v1",
  "approval_required": true,
  "expected_outcome": "carrier_assigned"
}
```

Business action event envelope

Use a versioned schema for all domain packs: event_id; occurred_at UTC; tenant_id; case_id; trace_id; span_id; agent_run_id; action_id; system; entity references; event_type; status; idempotency key; actor; redacted payload; evidence reference. Keep full sensitive payloads outside telemetry by default.

7. Integration and connector specifications

Customer systems supply the truth; the agent only proposes work.

Connector operating modes

Pattern	When used	Safety / implementation details
Read-only API	Discovery and shadow pilot	Scoped tokens; pagination; PII minimization; cache TTL
Stateful sandbox API	Demo and pre-production testing	Synthetic data; deterministic failures; reset fixtures
Trusted write adapter	Approved transactional work	Idempotency; compare-and-swap; audit; policy gate
Outbound webhook	New case, status changes	HMAC signature, replay window, deduplication
Polling connector	No event support	Incremental cursor, rate limits, retries
Customer-hosted bridge	Private network / restricted data	Outbound-only where possible; local secrets; selective export

Minimum connector interface

```
Connector.describe() -> capabilities, scopes, version
Connector.get(entity_type, external_id) -> record + version + observed_at
Connector.search(query, filters) -> records + provenance
Connector.propose(action) -> validated proposal + impact inputs
Connector.execute(approved_proposal, idempotency_key) -> receipt
Connector.verify(receipt, expected_state) -> observed_state + evidence
Connector.health() -> connectivity, latency, permissions
```

HTTP endpoints for OpsPilot core

Method / route	Purpose
POST /v1/cases	Create case; validate tenant, pack, objective
GET /v1/cases/{id}	Case state, evidence, timeline
POST /v1/cases/{id}/run	Start or resume workflow
POST /v1/cases/{id}/events	Insert domain events and signals
GET /v1/cases/{id}/proposals	Pending proposed business actions
POST /v1/approvals/{id}/decisions	Human approve/reject; version checked
POST /v1/cases/{id}/verify	Trigger authoritative reconciliation
POST /v1/demo/scenarios/{id}/reset	Idempotent fixture reset (demo only)
POST /v1/demo/scenarios/{id}/inject	Inject approved failure scenario (demo only)

Adapter onboarding checklist

- Map object IDs, owner system and system-of-record precedence.
- Inventory read/write operations and grant minimum privilege per tool.
- Define normal response, error response, partial commit, timeout and duplicate semantics.
- Specify change-data capture / webhook if available; choose polling otherwise.
- Establish idempotency and precondition support; if absent, design a safe customer-specific write wrapper.
- Define what constitutes authoritative proof of completion and how long it can take.
- Contract-test adapters against fixtures and fault injections.

8. Safety, policy, approvals and Blast Radius Lite

Agents may propose; trusted infrastructure decides whether an action can commit.

Action risk tiers

Tier	Examples	Minimum control
R0 — Read	Search case, fetch inventory	Scoped identity, retrieval audit
R1 — Draft	Prepare customer message or plan	Human preview for outbound content
R2 — Reversible write	Create internal follow-up task	Deterministic rule check, idempotency
R3 — External / costly write	Reassign carrier, issue purchase request	Fresh state, explicit limits, approval as configured
R4 — High consequence	Payment, legal denial, emergency dispatch	Exclude from demo autonomy; human-owned workflow

Blast Radius Lite

This is not an unlimited prediction engine. It traverses only explicitly modeled relationships: proposed action → directly affected business entity → related commitments and open transactions. Output includes known linked entities, possible downstream changes, data freshness, missing relationships and confidence caveats.

Domain	Example impact preview
Municipal	Affected service tickets, crew commitments, asset responsibility
Manufacturing	Other orders consuming stock; production slots; purchase cost
Logistics	Deliveries sharing vehicles; costs; service windows
Insurance	Claim tasks, repair estimates, required reviews; never coverage decisions
Property	Adjacent units, contractor slots, approvals, maintenance backlog

Approval and enforcement design

- Policy engine evaluates a normalized proposal; business context must be fetched and time-bounded.
- HOLD creates an approval task with evidence and expiry; no write is executed.
- ALLOW creates a short-lived execution authorization bound to proposal hash, actor and state version.
- Trusted adapters revalidate preconditions in the authoritative transaction where supported.
- DENY creates an auditable explanation. Agent may seek an alternative, never secretly bypass.
- Operational emergencies escalate to the organization's established response procedures.

OPA and Cedar can evaluate authorization/policies but do not themselves fetch business records, perform Blast Radius analysis or verify outcomes.

9. Evidence, observability and evaluation

A green trace is not necessarily a completed business outcome.

Three layers of evidence

Layer	Question	Typical source
Execution	Did the agent invoke a tool and receive a response?	OpenTelemetry span + execution receipt
Business state	Did the authoritative record actually change?	ERP / TMS / 311 / claims system query
Operational outcome	Was the task resolved at the intended business stage?	Acceptance event, completion evidence, reconciliation

Required measurements

- Task resolution rate and safe escalation rate (against a labeled test set).
- Correctness of selected tool and arguments; unsupported assertions and missing citations.
- Unauthorized write attempts, blocked actions, stale-version conflicts, duplicate attempts.
- Independent verification coverage and mismatch detection rate.
- Exception recovery success, replanning attempts, and human interventions per case.
- Run duration, model/tool costs, API latency, queue wait time and manual effort avoided.

Golden scenario evaluation suite

Scenario	Expected result
Happy path	Authorized outcome verified
Missing system data	Agent gathers alternate supported evidence or escalates
Tool 500 / timeout	State reconciled before any retry
Stale inventory or resource	No invalid write; replan required
Human rejects action	Action not executed; alternatives evaluated
Unexpected second actor	Concurrency control prevents conflicting commit
Prompt injection inside supplier email	Instructions treated as untrusted data; no policy override
Outcome not observed	Case marked PENDING/UNKNOWN, not success

For demos, seed at least 20 cases per domain: 8 straightforward, 8 exception-heavy, 4 deliberately adversarial. Maintain deterministic expected end-state labels and run regression evaluations on agent, prompt, tool or model changes.

10. Five industry domain packs

Business logic is configured once per vertical, while the orchestration remains shared.

CivicResolve — Citizen case coordination

Likely buyer: Municipal 311 or public-works operations lead

Dimension	Design detail
System adapters	311 CRM, asset registry, GIS jurisdiction lookup, crew dispatch, contractor queue, resident notification API
Read and write tools	search_cases; get_asset_owner; get_work_orders; get_crew_capacity; propose_reassignment; update_case; notify_resident
Deterministic guardrails	Jurisdiction and asset owner rules; human sign-off for priority changes; no automated legal/by-law decisions
Completion verifier	Accepted assignment + latest work-order status; physical completion only after technician evidence
Minimum synthetic fixtures	70 complaints, 25 assets, 12 crews, 15 work orders, overlapping tickets
Surprise to inject	Asset is owned by a utility, not by the municipality
Success KPI	Time-to-assignment; duplicate cases prevented; false closures

SupplyResolve — Production material recovery

Likely buyer: Plant COO or purchasing/operations manager

Dimension	Design detail
System adapters	ERP purchase orders, inventory across facilities, production schedule, supplier portal, purchasing approval
Read and write tools	get_supplier_delay; get_bom; check_inventory; get_open_pos; compare_approved_alternates; propose_transfer; create_purchase_request
Deterministic guardrails	Supplier approved; engineering substitute approval; spending thresholds; inventory version check
Completion verifier	Transfer / supplier acceptance + updated ATP quantity + schedule impact state
Minimum synthetic fixtures	50 SKUs, 6 suppliers, 15 work orders, 3 warehouses, 100 open POs
Surprise to inject	Preferred supplier rejects requested expedited shipment
Success KPI	Recovery cycle time; production commitments saved; exception decisions

FreightResolve — Shipment interruption recovery

Likely buyer: 3PL operations or dispatch director

Dimension	Design detail
System adapters	TMS, vehicle/telematics alerts, rate table, carrier availability, CRM, dispatch
Read and write tools	get_shipment; get_vehicle_alert; get_carriers; quote_alternative; reserve_carrier; update_dispatch; notify_customer
Deterministic guardrails	Carrier eligibility; cargo handling; carrier conflicts; approval above cost threshold
Completion verifier	Carrier acknowledgment + updated TMS assignment + continuing delivery status
Minimum synthetic fixtures	50 shipments, 18 trucks, 20 carriers, 10 rate contracts
Surprise to inject	Chosen carrier rejects assignment after plan is drafted
Success KPI	Time-to-recovery plan; accepted replacements; incremental cost

ClaimResolve — Claims file coordination

Likely buyer: Claims operations leader / TPA

Dimension	Design detail
System adapters	Claims system, policy records, repair-estimate portal, task queue, customer communications
Read and write tools	get_claim; get_required_docs; request_repair_update; compare_estimate_versions; create_adjuster_task; draft_status_update
Deterministic guardrails	No denial, fraud determination, coverage or payment approvals by the agent
Completion verifier	Required documents present + discrepancies recorded + adjuster-ready state, not settlement
Minimum synthetic fixtures	45 synthetic claims, 30 estimates, 14 repair facilities, policy references
Surprise to inject	New repair estimate contradicts prior inspection notes
Success KPI	Intake completeness; avoidable follow-ups; cycle-time to adjuster-ready

PropertyResolve — Tenant maintenance coordination

Likely buyer: Property manager or facilities director

Dimension	Design detail
System adapters	Property/tenant records, maintenance histories, contractor schedules, work orders, communications

Dimension	Design detail
Read and write tools	get_property; classify_safety_flags; check_contractors; request_appointment; create_work_order; update_resident; verify_completion
Deterministic guardrails	Immediate human escalation for emergencies; access and spending rules; tenant privacy
Completion verifier	Vendor acceptance + owner action + completion evidence; do not infer physical repair
Minimum synthetic fixtures	60 synthetic units, 15 contractors, 35 work orders, 8 emergencies
Surprise to inject	Contractor declines and adjacent unit reports similar leak
Success KPI	Time-to-assignment; missed appointments; unresolved repair backlog

11. Detailed demo scenarios and fault injection

Demo every industry with an outcome, not an animated scripted conversation.

Industry	Opening line the buyer understands	Mid-demo surprise	Demonstrated safe conclusion
Municipal	This streetlight complaint has not moved in three days.	Original crew unavailable; utility jurisdiction found	Correct owner accepts case, resident receives verified update
Manufacturing	Friday shipment is at risk because a component is late.	Supplier refuses expedited option	Alternative transfer/purchase proposal approved; ATP checked
Logistics	A truck with three urgent deliveries breaks down.	Replacement carrier withdraws	New eligible carrier accepts, TMS reflects reassignment
Insurance	This automobile claim has waited for a usable estimate.	Updated estimate contains conflicting figures	Targeted follow-up generated; adjuster-ready only with evidence
Property	Tenant reports an active water leak.	Vendor declines; adjoining apartment affected	Urgent escalation, alternative dispatch and progress tracking

Universal eight-minute live demo script

- 0:00–0:45 — Show the human business problem and the cost of leaving it unresolved.
- 0:45–2:15 — Start a case; show investigation across three business systems.
- 2:15–3:30 — Present an agent-generated plan with source evidence and alternatives.
- 3:30–4:30 — Inject one plausible complication; observe re-planning.
- 4:30–5:45 — Show policy hold, human approval or safe escalation.
- 5:45–7:15 — Execute a guarded write and query the system of record.
- 7:15–8:00 — Show business result, audit evidence and KPI baseline.

The demo control panel must include

Start scenario; choose industry; run in guided or autonomous mode; inject exception; pause or resume; view tool execution; approve/reject; reset to a known fixture; export case evidence. The sales presenter must be able to perform a complete demo without database editing or developer intervention.

12. User experience and operator screens

Build one interface that feels familiar to an operations team.

Screen	Must show	User action
Industry launcher	Five domain tiles, sample business problem, estimated demo time	Select / start scenario
Case inbox	Priority, owner, due status, pending approvals, blockers	Open / assign / escalate
Case workspace	Objective, known facts, source systems, next action, timeline	Review evidence
Plan + impact	Proposed action, alternatives, related entities, assumptions	Approve / reject / request changes
Execution trail	Tool outcomes, version checks, idempotency receipt	Inspect why something happened
Outcome verification	Expected vs observed state, timestamp, proof, missing evidence	Recheck / close / escalate
Demo studio	Fault injectors, synthetic fixture versions, reset and replay	Guide demonstration
Ops & evaluation	Completion / escalations / latency / failures / model costs	Filter, compare releases

Usability standards

- Use plain operational language such as “carrier accepted” rather than “tool response 200”.
- Display pending / unknown separately from failed and completed.
- Make approval decisions comprehensible: action, affected entity, cost, risk, expiry and evidence.
- Show direct source records and timestamps for important facts.
- Keyboard-accessible UI, readable contrast, screen reader semantics and clear error recovery.

13. Repository and environment architecture

Separate platform runtime, domain rules and connectors for maintainability.

Suggested monorepo structure

```
opspilot/
  apps/web/              # Next.js sales/operator UI
  services/api/          # FastAPI tenant and case endpoints
  services/agent-runtime/   # LangGraph workers + checkpoints
  services/policy/       # rules + approvals + impact evaluation
  services/simulator/    # stateful mock business applications
  packages/contracts/    # Pydantic and TS API contracts
  packages/connectors/   # common adapters + auth / retries
  domain-packs/civic/    # tools, fixtures, invariants, verifier
  domain-packs/supply/
  domain-packs/freight/
  domain-packs/claims/
  domain-packs/property/
  tests/scenarios/       # happy, fault, concurrency, red-team
  infra/docker/          # compose, migrations, seed data
  docs/                  # ADRs, runbooks, demo scripts
```

Recommended implementation defaults

Concern	Choice / reasoning
Runtime	Python FastAPI and LangGraph for durable agent transitions
Frontend	Next.js, Tailwind; one design system and domain-themed content
Primary database	PostgreSQL with relational transaction and event tables
Background processing	Job queue or durable workflow engine; no untracked in-process background work
Schemas	Pydantic and OpenAPI-generated TypeScript types
Observability	OpenTelemetry spans for model/tool execution, business ledger for outcome evidence
Deployment	Docker Compose local; containers in customer-approved cloud for pilots
LLMs	Provider adapter with configurable model, output schema, timeouts and budgets

Engineering guardrails

Pin dependencies, version prompts/tools, manage migrations through Alembic, use feature flags for domain packs, keep secrets in a managed vault, and enforce CI gates for unit, integration, scenario and adversarial tests. Separate demo data stores from any future customer systems.

14. Security, privacy and deployment readiness

Demonstrate enterprise discipline before asking for real data or credentials.

Minimum security controls

- OIDC authentication; role-based permissions; optionally customer SSO for pilots.
- Tenant isolation on every case, tool, business entity and evidence record.
- Least privilege connectors, separate read and write credentials, key rotation and audit.
- No unrestricted internet, shell or file-system actions available to the agents.
- Prompt-injection defenses at untrusted documents, search responses and tool outputs.
- Explicit region and retention settings; PII redaction in telemetry; encrypted storage and transport.
- Exportable audit evidence, incident process, backups, recovery objectives and operational runbooks.

Canada-specific considerations

Treat Canadian privacy and public-sector requirements as project-specific. PIPEDA cross-border processing does not universally prohibit overseas processors, but organizations retain accountability and must address the arrangement through disclosure, contractual controls and monitoring. Municipal requirements vary by jurisdiction and procurement. Ontario's Responsible Use of AI Directive applies to Ontario ministries and provincial agencies; do not automatically assume it governs municipalities in identical form. Public-sector designs should nonetheless consider transparency, human oversight, accessibility, records retention and appeal rights.

Deployment disclaimer

The synthetic demonstration does not establish compliance with PIPEDA, provincial health laws, municipal procurement, AODA or any customer's security requirements. Conduct legal, privacy and security reviews before production work.

Customer-hosted pilot pattern

For sensitive clients, keep credentials and raw PII inside the customer's environment. Install a minimal connector that resolves business facts and executes approved actions, with the OpsPilot controller receiving only necessary redacted metadata, correlation IDs and evidence references. Define data flow and responsibility in the statement of work.

15. Positioning and marketing strategy

Sell operations resolved, not orchestration frameworks.

Positioning statement

Prudlelabs OpsPilot helps Canadian organizations resolve operational exceptions using AI agents that investigate across existing systems, coordinate approved actions, adapt to disruption and verify progress — without replacing core business software.

Value proposition by buyer

Buyer	Pain to lead with	Demo story	Proof to request
Municipal operations	Backlog, handoffs, resident follow-up	Repeated public-service complaint	Time-to-assignment and false closures
Plant COO	Material shortages threaten orders	Delayed component with alternatives	Time-to-recovery and avoided rescheduling
Logistics director	Dispatch exceptions overwhelm staff	Breakdown and backup carrier	Exception resolution time and carrier cost
Claims manager	Claims stall on missing information	Incomplete estimate and follow-ups	Adjuster-ready time
Property operator	Contractor scheduling and tenant complaints	Leak, refusal and escalation	Accepted dispatch and rework

Messaging hierarchy

- Headline: From operational problem to verified action.
- Subhead: Agents that investigate, plan, coordinate and follow through inside your existing systems.
- Proof: Watch a realistic problem change mid-demo. See what the agent does next.
- Trust: Approval gates, traceable decisions, authoritative-state checks and escalation.
- CTA: Book a 20-minute industry demonstration or a paid workflow discovery workshop.

Avoid these claims

Do not promise full autonomy, zero errors, guaranteed cost reduction, human replacement, universal integrations, automatic regulatory compliance, or 100% Blast Radius prediction. All ROI claims need measurement in an agreed pilot baseline.

16. Sales assets and demo-to-pilot motion

One buyer story, one working demo and one measurable next step.

Minimum content kit

Asset	Purpose	Format
Industry one-pager × 5	Recognizable problem, before/after, systems, result	PDF / webpage
Three-minute demo video × 5	Show investigation, surprise and outcome	Captioned video
Executive overview deck	OpsPilot thesis, credibility, industry coverage	8–10 slides
Technical architecture sheet	Security, adapters, approvals, evidence	2–3 pages
Pilot discovery questionnaire	Process, constraints, systems, baseline KPI	Editable form
Security / privacy factsheet	Deployment, data flow, retention, access controls	2 pages
Pilot SOW template	Scope, integrations, acceptance, roles and support	DOCX
ROI worksheet	Time saved and quality impacts with baseline	Spreadsheet

Sales conversation flow

- Open with the relevant operational incident, not a technology overview.
- Ask its frequency, current owner, system touchpoints and delays.
- Show an unexpected event and evidence-backed recovery.
- Clarify permitted actions and required human approvals.
- Confirm API access, past cases and one measurable KPI.
- Offer paid discovery and a bounded pilot with success criteria.

Typical buyer objections

Already have Copilot? Keep it; use OpsPilot only where cross-system exceptions add value.

Wrong decision? Protected writes require policy, evidence and approvals.

No production access? Use synthetic data first, then customer-hosted read-only connectors.

Scripted demo? Inject a new event and inspect the changed tool calls and state.

Where is ROI? Compare a controlled pilot with a measured manual baseline.

17. Packaging, pricing and partnerships

Use service-led commercialization before committing to a SaaS price book.

Commercial offers

Offer	Deliverables	Indicative CAD fee / status
Industry demo / workshop	Live scenario, business process discussion, fit check	No-charge selective demo; 20–30 minutes
Paid workflow assessment	Process map, integrations, risk profile, ROI baseline, pilot plan	C$4k–10k hypothesis
Controlled pilot	One workflow, 1–2 systems, approvals, evidence, supervised evaluation	C$20k–55k hypothesis
Production integration	Customer connectors, security, deployment, testing, change management	Custom SOW; start from scoped discovery
Managed operations	Monitoring, updates, evaluation, incident support	Retainer quoted by environment and SLA

Prices above are proposed commercial experiments, not market quotes or guarantees. Validate through conversations with Canadian buyers and local delivery costs.

Potential partner categories

- Canadian managed service providers and Microsoft 365 / Dynamics consultants.
- ERP integrators serving manufacturing and distribution.
- TMS implementation partners and transport technology vendors.
- Municipal software and public sector systems integrators.
- Industry associations, innovation centres and qualified AI adoption advisers.

Funding-aware selling

FedDev Ontario's Regional Artificial Intelligence Initiative currently has application windows and defined eligibility criteria. Prudlelabs could participate as an implementation provider on a qualifying customer-led project, subject to the applicant's eligibility and approved budget. Never market funding as guaranteed or describe it as a grant directly owed to Prudlelabs.

18. Canada 90-day business development plan

Use existing Ontario relationships to test repeatable demand, not only bespoke requests.

Window	Business development activity	Leading indicator
Days 1–15	Package CivicResolve and FreightResolve demos; prepare case story, demo videos, one-pagers and technical assurance sheet.	Two tested demos and two vertical landing pages
Days 16–30	Identify 30 Ontario buyers/partners; book industry-specific meetings; map procurement calendars and integration partners.	10 discovery conversations targeted
Days 31–60	Run live demos, capture objection patterns, offer scoped paid assessments and one reference partnership.	2 paid assessments targeted
Days 61–90	Convert first scoped pilot, capture baseline metrics, create approved anonymous case study.	1 signed pilot targeted

Reference and credibility plan

If Prudlelabs can disclose the Brampton municipal engagement, publish only information expressly permitted by the agreement and client. Clearly separate an awarded project from delivered results; do not imply endorsement by the city or claim a specific AI outcome without evidence.

Channels and campaigns

- Industry micro-webinars: “What happens when your usual operations automation encounters an exception?”
- Targeted LinkedIn outreach to operations leadership using one industry scenario per campaign.
- Co-host practical AI operations workshops with ERP/MSP/municipal technology partners.
- Publish short comparison videos: fixed automation versus agentic exception handling, with measured limits.
- Use a lightweight discovery checklist as the lead magnet; follow with demo and paid assessment.

19. ROI measurement and success criteria

Prove customer value without invented savings.

Pilot baseline and measurement design

Metric	Baseline measurement	Pilot measurement
Case processing time	Median time across comparable manually handled cases	Median end-to-end agent+human time
Human touch time	Minutes of active staff work per case	Minutes of active approval/exception work
Completion quality	Expert-audited correctly resolved cases	Verified resolution against labeled outcomes
Rework / duplication	Repeated updates, incorrect work orders, duplicate records	Observed rework and prevented duplicates
Customer communication	Time to accurate customer update	Time to verified status update
Operational cost	Staff time and externally invoiced recovery cost	AI/API cost + staff time + implementation allocation

Illustrative ROI calculation (not a forecast)

```
Monthly benefit = (cases × minutes saved per case / 60 × loaded hourly cost)
                + verified error/rework costs avoided
Monthly net benefit = monthly benefit - AI / hosting / support cost
Payback months = initial project cost / positive monthly net benefit
```

Separate realized savings, estimated capacity released, and unverified avoided exposure. Only count error prevention where there is a validated baseline and an accepted attribution method.

Go/no-go gates

- Buyer: named sponsor, accountable process owner and realistic integration access.
- Technical: at least 90% of labeled safe cases reach a correct outcome or safe escalation in demo test suite (target, not observed result).
- Safety: zero unauthorized critical writes in adversarial and concurrency tests.
- Commercial: credible willingness to fund integration and measurable follow-up work.
- Maintainability: common engine remains shared; no copy-pasted fork per domain.

20. 17-week engineering implementation plan

Build one robust execution spine before expanding the number of dashboards.

Weeks	Delivery	Definition of done
1–2	Foundation: monorepo, auth, schemas, case data, CI/CD	Case CRUD, migrations, seeded users, basic UI
3	LangGraph runtime, tools, receipts, pause/resume	One case survives process restart; action ledger complete
4–5	Freight simulator and happy-path workflow	Agent can select tools and verify stateful dispatch update
6	Freight exceptions, approval, concurrency	Replans carrier rejection and blocks stale / duplicate commits
7–8	Civic domain pack and demo controller	Municipal cross-system case with jurisdiction surprise
9–10	Supply domain pack	Supplier decline and safe material recovery tested
11–12	Property domain pack	Urgent escalation, vendor rejection and follow-up
13–14	Claims domain pack	Estimate discrepancy and adjuster handoff
15	Blast Radius Lite and domain verifiers	Impact preview evidence + known limitations displayed
16	Evaluation, prompt injection, reliability and audit	Labeled scenario suite and release gates
17	Sales packaging, QA, record videos and release	Five resettable demos and runbooks

Staffing planning assumptions

Minimum: one backend/agent engineer, one backend/integration engineer, one frontend engineer, part-time QA and product/domain reviewer. The sequence assumes shared full-time engineering capacity and concurrent UX/content support; it is a planning estimate rather than a fixed delivery commitment.

21. Development epics, backlog and tests

Ready for conversion into GitHub issues, Azure Boards or Jira.

Epic	Priority	Acceptance criteria
E01: Tenant, user, identity	P0	Tenant-isolated API, role checks, audit events
E02: Case lifecycle	P0	Persist/resume, versions, owner, timeline
E03: Tool registry + contracts	P0	Typed input/output, scope and failure classes
E04: Agent planning	P0	Evidence-backed plan with bounded tool selections
E05: Approval & guards	P0	No protected write without valid approval/version
E06: Connector simulator	P0	Stateful reads/writes; deterministic seeded cases
E07: Outcome verifier	P0	Actual state fetched independently; mismatch flagged
E08: Demo injector/reset	P0	Named faults can be replayed reliably
E09: Industry packs	P1	Each pack meets domain scenario and verifier tests
E10: Impact preview	P1	Correct linked entities and freshness caveats
E11: Evaluation/telemetry	P0	Trace, business event, labeled golden tests
E12: Pilot hardening	P2	Secrets, SSO, retention, security review, backups

Release test matrix

- Unit: authorization rules, state transitions, extraction schema, amount/time arithmetic.
- Contract: tool schema validation, external ID mapping, error parsing and tenant boundaries.
- Integration: connector adapters with transactional simulator and injected network faults.
- E2E: operator creates case, views plan, rejects/approves, verifies result, exports evidence.
- Concurrency: two cases attempt same carrier or stock reservation; only valid allocation persists.
- Security: prompt injection, forged approvals, replayed webhook, cross-tenant access, tampered state version.
- Regression: replay the full golden scenario suite on prompt/model/tool changes.

22. Risks, decisions, glossary and sources

Consolidated design assumptions and external references.

Decision register

Decision	Recommended position
Brand	Prudlelabs OpsPilot. Use the same spelling across UI, docs, campaigns and code.
Architecture	Shared runtime with five domain packs; do not build five isolated applications.
Model approach	Provider-independent LLM interface; no mandatory multi-agent architecture.
Writes	Deterministic policy + versioned approval + trusted execution adapter.
Proof	Always verify actual business state; pending is not success.
Data	Synthetic-only demo; customer-specific connector/privacy assessment for pilot.
Product route	Sell implementations first; validate reusable SaaS and Blast Radius later.

Known delivery risks

- Demo mistaken for production proof — label synthetic systems and present clear limitations.
- LLM variability — use fixture assertions, strict schemas and bounded model settings.
- Integration complexity — qualify API access and data owner before committing fixed prices.
- Unsafe autonomy — enforce protected writes and emergency escalations outside model control.
- Domain misunderstanding — validate each pack with a Canadian domain reviewer before public sales.
- Fragmented codebase — make every domain pack use shared event, action, approval and verifier contracts.

Short glossary

Term	Meaning
Case	One tracked business problem and its evolving resolution state
Domain pack	Industry-specific tools, entities, rules, fixtures and verifiers
Business invariant	A rule that must remain true across a business transaction
Idempotency key	A unique identity used to avoid duplicate business writes
Authoritative system	System whose committed state determines the business truth
Blast Radius Lite	Scoped preview of related entities affected by a proposed action

Term	Meaning
Safe escalation	Handing a case to a human with evidence rather than fabricating completion

Sources and documentation — reviewed 20 October 2026

Statistics Canada — AI use by Canadian businesses, Q2 2026
https://www150.statcan.gc.ca/n1/pub/11-621-m/11-621-m2026010-eng.htm

City of Brampton — AI Proof of Concept Program
https://www.brampton.ca/EN/City-Hall/AI-POC/

FedDev Ontario — RAII business eligibility
https://feddev-ontario.canada.ca/en/funding-southern-ontario/regional-artificial-intelligence-initiative-businesses-who-can-apply

FedDev Ontario — RAII application window
https://feddev-ontario.canada.ca/en/funding-southern-ontario/regional-artificial-intelligence-initiative-businesses-when-apply

Ontario — Responsible Use of Artificial Intelligence Directive
https://www.ontario.ca/page/responsible-use-artificial-intelligence-directive

Ontario — Building accessible government services
https://www.ontario.ca/page/making-government-services-accessible

Office of the Privacy Commissioner — Cross-border processor accountability
https://www.priv.gc.ca/en/opc-actions-and-decisions/investigations/investigations-into-businesses/2020/pipeda-2020-001/

LangGraph — Persistence and checkpointing
https://langchain-ai.github.io/langgraph/concepts/durable_execution/

OpenTelemetry — Agent and workflow semantic conventions (development)
https://github.com/open-telemetry/semantic-conventions-genai/blob/main/docs/gen-ai/gen-ai-agent-spans.md

Cedar — Authorization policy language reference
https://docs.cedarpolicy.com/

The first release: an operational demo, not another AI slide deck

A narrow, working success is worth more than five conceptual interfaces.

The recommended first milestone is a complete FreightResolve case: detect truck breakdown, investigate shipment and carrier systems, propose alternatives, inject a carrier rejection, require manager approval, execute a valid reassignment, and verify acceptance in the simulated TMS. The same engine should then run the CivicResolve scenario without duplication of orchestration logic.

Approval requested from Prudlelabs leadership

Authorize the shared engine plus two reference domain packs as the initial engineering scope. Keep the remaining three domain packs as sequenced extensions. Validate target buyer demand and integration access in parallel with development.

Branding throughout this document: Prudlelabs OpsPilot.